

This is a particular case of a *similarity transform* of the matrix $\mathbf{A}$,

$$\mathbf{A} \rightarrow \mathbf{Z}^{-1} \cdot \mathbf{A} \cdot \mathbf{Z} \quad (11.0.12)$$

for some transformation matrix $\mathbf{Z}$. Similarity transformations play a crucial role in the computation of eigenvalues, because they leave the eigenvalues of a matrix unchanged. This is easily seen from

$$\begin{aligned} \det |\mathbf{Z}^{-1} \cdot \mathbf{A} \cdot \mathbf{Z} - \lambda \mathbf{1}| &= \det |\mathbf{Z}^{-1} \cdot (\mathbf{A} - \lambda \mathbf{1}) \cdot \mathbf{Z}| \\ &= \det |\mathbf{Z}| \det |\mathbf{A} - \lambda \mathbf{1}| \det |\mathbf{Z}^{-1}| \\ &= \det |\mathbf{A} - \lambda \mathbf{1}| \end{aligned} \quad (11.0.13)$$

Equation (11.0.11) shows that any matrix with complete eigenvectors (which includes all normal matrices and “most” random nonnormal ones) can be diagonalized by a similarity transformation, that the columns of the transformation matrix that effects the diagonalization are the right eigenvectors, and that the rows of its inverse are the left eigenvectors.

For real symmetric matrices, the eigenvectors are real and orthonormal, so the transformation matrix is orthogonal. The similarity transformation is then also an *orthogonal transformation* of the form

$$\mathbf{A} \rightarrow \mathbf{Z}^T \cdot \mathbf{A} \cdot \mathbf{Z} \quad (11.0.14)$$

While real nonsymmetric matrices can be diagonalized in their usual case of complete eigenvectors, the transformation matrix is not necessarily real. It turns out, however, that a real similarity transformation can “almost” do the job. It can reduce the matrix down to a form with little two-by-two blocks along the diagonal and all other elements zero. Each two-by-two block corresponds to a complex-conjugate pair of complex eigenvalues. We will see this idea exploited in some routines given later in the chapter.

The “grand strategy” of virtually all modern eigensystem routines is to nudge the matrix $\mathbf{A}$ toward diagonal form by a sequence of similarity transformations,

$$\begin{aligned} \mathbf{A} &\rightarrow \mathbf{P}_1^{-1} \cdot \mathbf{A} \cdot \mathbf{P}_1 \rightarrow \mathbf{P}_2^{-1} \cdot \mathbf{P}_1^{-1} \cdot \mathbf{A} \cdot \mathbf{P}_1 \cdot \mathbf{P}_2 \\ &\rightarrow \mathbf{P}_3^{-1} \cdot \mathbf{P}_2^{-1} \cdot \mathbf{P}_1^{-1} \cdot \mathbf{A} \cdot \mathbf{P}_1 \cdot \mathbf{P}_2 \cdot \mathbf{P}_3 \rightarrow \text{etc.} \end{aligned} \quad (11.0.15)$$

If we get all the way to diagonal form, then the eigenvectors are the columns of the accumulated transformation

$$\mathbf{X}_R = \mathbf{P}_1 \cdot \mathbf{P}_2 \cdot \mathbf{P}_3 \cdot \dots \quad (11.0.16)$$

Sometimes we do not want to go all the way to diagonal form. For example, if we are interested only in eigenvalues, not eigenvectors, it is enough to transform the matrix $\mathbf{A}$ to be triangular, with all elements below (or above) the diagonal zero. In this case the diagonal elements are already the eigenvalues, as you can see by mentally evaluating (11.0.2) using expansion by minors.

There are two rather different sets of techniques for implementing the grand strategy (11.0.15). It turns out that they work rather well in combination, so most modern eigensystem routines use both. The first set of techniques constructs individual $\mathbf{P}_i$ ’s as explicit “atomic” transformations designed to perform specific tasks,